L4: The Role of the Exponent of a Power-law Distribution

What is the mean and the variance of a power-law degree distribution?

More generally we can ask: what is the m'th statistical moment of a power-law degree distribution?

It is defined as:

$$E[k^m] = \sum_{k_{min}}^{\infty} k^m p_k = c \sum_{k_{min}}^{\infty} k^{m-\alpha}$$

where **c** is the proportionality coefficient we derived in the previous page.

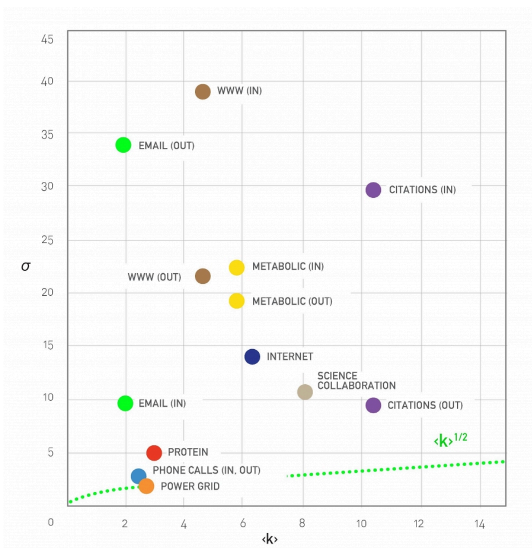
If we rely again on the continuous k approximation, the previous summation becomes an integral that we can easily calculate:

$$E[k^m] = c \int_{k_{min}}^{\infty} k^{m-\alpha} dk = \frac{c}{m-\alpha+1} k^{m-\alpha+1} |_{k_{min}}^{\infty}$$

Note that this integral diverges to infinity if $m - \alpha + 1 \geq 0$ and so, the m'th moment of a power-law degree distribution is well defined (finite) if $m < \alpha - 1$.

Consequently, the mean (first moment) exists if $\alpha > 2$ and the variance (second moment minus the square of the mean) exists if $\alpha > 3$.

Of course the variance cannot be "infinite" if the network has a finite number of nodes (i.e., k never "extends to infinity" in real networks). For many real-world networks however, the exponent α is estimated to be between 2 and 3, which means that even though the distribution has a well-defined average degree, the variability of the degree across different nodes is extremely large.



Standard Deviation is Large in Real Networks, Figure 4.8 from networksciencebook.com by Albert-László Barabási

To illustrate this last point, let's look at the relation between the average degree and the standard deviation of the degree distribution for several real-networks (for more details about these networks please review Table 4.1 of your textbook).

For a network with Poisson degree distribution (such as random ER graphs), the degree variance is equal to the average degree $\left(E[k] = \bar{k} = \sigma^2\right)$ and so $\sigma = \sqrt{\bar{k}}$

Note that many real-world networks have much higher σ than that -- in some cases σ is even an order of magnitude larger than $\bar{k}$. In the case of the WWW in-degree distribution, for example, the average in-degree is only around 4 while the standard deviation of the in-degree is almost 40!

Food for Thought

Prompt: repeat the derivation for the m'th moment outlined above in more detail.